

Beyond Detection Parity: Localization Parity in Pedestrian Detection

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Abstract. Pedestrian detection is a safety-critical component of autonomous driving and intelligent surveillance, where equitable performance across demographic groups is essential for trustworthy deployment. Fairness research in this domain has evaluated performance almost exclusively through detection recall, that is, whether a person is found at all, leaving the precision of the predicted bounding box unexamined. Because existing audits reduce localization to a binary hit-or-miss threshold, systematic disparities in how tightly detectors localize different groups remain invisible, even though loose boxes directly compromise downstream distance and trajectory estimates. This paper introduces *Localization Parity*, a fairness criterion that requires equal expected Intersection over Union across demographic groups conditional on detection. We present the first confounder-controlled audit spanning four driving datasets (BDD100K, CityPersons, EuroCity-Day, and EuroCity-Night), three detector families (YOLOX, RetinaNet, and Cascade R-CNN), and two demographic axes (gender and age), together with a demographic-weighted bounding box regression intervention. Female and child pedestrians receive significantly looser boxes in all twelve dataset and detector combinations, with mean gaps of 1.40 and 3.27 percentage points. These gaps survive scale stratification in 35 of 36 and 24 of 30 cells, respectively. The proposed intervention closes the training-set gap by 46.8%, yet fails to generalize to validation data. Localization Parity therefore exposes a previously unmeasured and safety-relevant axis of algorithmic unfairness, and the resulting fairness generalization gap poses a concrete open problem for equitable perception systems.

Keywords: Localization Parity · Algorithmic Fairness · Pedestrian Detection · Bounding Box Regression · Demographic Bias · Object Detection

1 Introduction

Pedestrian detection is central to modern computer vision pipelines for self-driving cars, surveillance, and traffic infrastructure. The cost of error is not shared evenly across demographic groups. Prior work has shown that *detection*

recall is unequal: Wilson et al. [14] report lower average precision for darker-skinned pedestrians, while Brandao [2] and Li et al. [9] find substantially higher miss rates for children than for adults across many detectors.

What these audits leave unmeasured is *localization quality*. Once a detector draws a box, the tightness of that box shapes every estimate that follows: position, distance to the ego vehicle, predicted trajectory, and braking urgency. Two detections can both pass the conventional $\text{IoU} \geq 0.5$ hit threshold and still differ sharply in geometry. A box at IoU 0.51 and a box at IoU 0.91 look identical under recall-based fairness metrics, yet they feed very different signals into downstream modules. If one demographic group systematically receives looser boxes, that group systematically receives worse downstream estimates, and no current fairness benchmark would notice.

We formalize this overlooked criterion as **Localization Parity (LP)**: equality of expected Intersection over Union (IoU) across demographic groups, conditional on successful detection and after controlling for confounders such as object scale. We then run the first multi-dataset, multi-detector LP audit and test a training-time mitigation.

Contributions.

1. A formal definition of Localization Parity as a group fairness criterion on box regression quality, orthogonal to detection parity, with a clear link to existing fairness notions for classification and regression (Section 3).
2. The first confounder-controlled LP audit across four autonomous driving datasets, three detector families spanning single-stage and two-stage designs, and two demographic axes: *gender* (female vs. male) and *age* (child vs. adult). Race and skin tone are not audited in this study (Section 6.1).
3. Empirical evidence that female and child pedestrians receive consistently lower IoU boxes (mean gaps -1.40 pp and -3.27 pp), surviving ICON² scale stratification [13].
4. Demographic-Weighted Bounding Box Regression (DWBBR), which closes the training gap by 46.8% yet fails to transfer cleanly to held-out data. We document this *fairness generalization gap* as an open problem.

Research questions. Three questions organize the empirical study:

- RQ1** Does a Localization Parity gap exist across demographic groups in mainstream pedestrian detectors? Concretely, do female and child pedestrians receive lower mean IoU than male and adult pedestrians across datasets and architectures?
- RQ2** If a gap exists, does it survive confounder control for object scale (and occlusion where annotated)? Scale is the critical check: if the gap is only “children are small,” it is a size problem, not a fairness one.
- RQ3** Can a targeted training-time intervention shrink the gap, and does any improvement transfer to held-out data?

2 Related Work

Fairness in computer vision. Group fairness (demographic parity and equalized odds) [7], individual fairness, and calibration form the classical toolkit. Vision audits usually operationalize group fairness through equal detection rates or equal average precision. Wilson et al. [14] established predictive inequity across Fitzpatrick skin-tone groups on BDD100K using AP. Schumann et al. [12] critique binary third-party gender labels and argue that annotation protocols themselves can encode bias. Fair regression definitions [1] constrain disparities in continuous prediction error; LP applies the same idea to the IoU of matched detections (Section 3). What remains missing in this literature is a criterion that isolates box geometry after a detection has already been accepted.

Pedestrian detection and demographic bias. BDD100K [15], CityPersons [16], and EuroCity Persons [3] are the dominant driving benchmarks. Li et al. [9] and Brandao [2] show elevated child miss rates across many detectors, and Li et al. further report higher female miss rates under low light. Khoshkdahan et al. [8] study pose and occlusion as structural confounders of detection accuracy. All of these works evaluate hit-or-miss outcomes rather than continuous localization quality. Cascade refinement [4] and focal reweighting [10] improve overall box precision, but neither was designed to equalize IoU across demographic groups.

Confounder control and fair training. ICON² [13] stratifies by scale and occlusion so that demographic gaps cannot be attributed to size alone; we apply it to IoU rather than AP. Loss reweighting and Group DRO [11] are standard tools for fair training, but they have been used mainly for classification or recall rather than for the bounding box regression branch. Focal Loss [10] reweights hard classification examples; DWBBR instead reweights the CIoU regression term by demographic group performance. To our knowledge, no prior work both audits continuous localization fairness across a multi-dataset detector matrix and tests whether regression reweighting closes that gap out of sample.

3 Localization Parity

3.1 Formal Definition

Let $\mathcal{D} = \{(b_i, \hat{g}_i, g_i)\}$ be the set of successfully detected instances, where b_i is the predicted box, $\hat{g}_i$ the ground-truth box, and $g_i \in \{g_1, \dots, g_k\}$ the demographic group of instance i . Matching follows the COCO convention: a prediction counts as a detection only if $\text{IoU}(b_i, \hat{g}_i) \geq 0.5$. Localization Parity requires

$$\mathbb{E}[\text{IoU}(b_i, \hat{g}_i) \mid g_i = g_a] = \mathbb{E}[\text{IoU}(b_i, \hat{g}_i) \mid g_i = g_b] \quad (1)$$

for every pair of groups $a \neq b$, after controlling for instance scale. We report the signed gap

$$\Delta\text{IoU} = \mathbb{E}[\text{IoU} \mid g_{\text{priv}}] - \mathbb{E}[\text{IoU} \mid g_{\text{dis}}],$$

where g_{priv} denotes the privileged group (male, adult) and g_{dis} the disadvantaged group (female, child). A negative gap means the disadvantaged group receives worse boxes.

LP is deliberately conditional on detection. It equalizes box quality among people who were found, and is therefore orthogonal to detection parity (equal recall or miss rate). A detector can satisfy one while violating the other; both are needed for safety-relevant fairness.

3.2 Relation to Existing Fairness Criteria

LP sits at the intersection of group fairness and fair regression:

- Relative to demographic parity and equalized odds for classification [7, 5], LP replaces a binary outcome (detected or not) with a continuous quality score (IoU), evaluated only on the detected set. This is analogous to equalized opportunity applied to localization quality.
- Relative to fair regression [1], which seeks bounded disparity in expected loss across groups, LP uses the loss $\ell = 1 - \text{IoU}$ on matched boxes.
- Relative to AP-based audits [14], LP isolates the localization component that AP mixes with classification confidence and ranking.

LP does not claim individual fairness or calibration. It is a group-level criterion because the safety harm of interest, systematically worse downstream estimates for a demographic group, is itself group structured.

3.3 Why Small IoU Gaps Matter Downstream

A natural question is whether gaps of 1–3 pp IoU are practically consequential. Under a pinhole projection, relative depth error scales with relative box-height error. For a pedestrian at 30 m whose true box height is 80 px, a 2 pp mean IoU deficit (roughly 2–4 px of systematic height or vertical-center error) translates into about 0.7–1.5 m of depth bias. That is enough to alter time-to-collision estimates at urban speeds and to shift predicted crossing trajectories. Because trajectory and collision modules are typically nonlinear in box geometry, group-level mean gaps of this magnitude can compound across the planning stack even when every detection still clears the 0.5 IoU threshold. We therefore treat LP gaps in this range as safety-relevant, not merely statistically detectable. A full end-to-end propagation study remains future work (Section 8).

4 Experimental Setup

4.1 Datasets

We evaluate on four pedestrian benchmarks chosen for geographic, sensor, and lighting diversity:

- **BDD100K** [15]: large-scale US dashcam data with about 100,000 annotated pedestrian boxes and binary male/female labels.

- **CityPersons** [16]: about 19,000 boxes over 2,975 Cityscapes street-level images, with careful occlusion annotation.
- **EuroCity-Day** and **EuroCity-Night** [3]: subsets of EuroCity Persons (about 238,200 boxes over 47,300 images from 31 European cities), split by lighting.

Demographic labels for CityPersons and EuroCity use augmented annotations where available. Detections with missing or ambiguous labels are dropped. Label quality limitations are discussed in Section 7.

4.2 Detectors

We cover the three main detector design families with public pretrained weights and no fine-tuning in the audit phase:

- **YOLOX** [6]: single-stage, anchor-free;
- **RetinaNet** [10]: single-stage, anchor-based;
- **Cascade R-CNN** [4]: two-stage, with cascaded IoU refinement aimed at tighter localization.

The mitigation experiments use **YOLOv8n** (Ultralytics), the lightweight YOLO variant that serves as our trainable single-stage backbone.

4.3 Demographic Axes

The two protected attributes studied are:

1. **Gender:** female vs. male;
2. **Age:** child vs. adult.

Race and skin tone are not among the audited axes. An earlier pilot examined skin tone, but the labeled sample was too small and the categories too coarse for reliable statistical claims. We therefore exclude it here and mark it as future work (Section 7).

4.4 Audit Protocol and Confounder Control

Matching. Predictions are matched to ground truth by greedy, confidence-sorted IoU assignment. Matches with $\text{IoU} \geq 0.5$ enter the LP analysis; false positives are excluded.

Confounders. The primary confounder is **object scale** (box height), because female and child pedestrians tend to occupy smaller image regions, and small objects are harder to localize for any reason. Where annotations permit, we also consider **occlusion**. Following ICON² [13], we bin ground-truth instances within each dataset into short (≤ 33 rd height percentile), medium (34th–66th), and tall (≥ 67 th) strata, then recompute per-group mean IoU inside each bin. A gap that remains significant within strata cannot be attributed to scale alone.

Statistical testing. Within each cell and each stratum we compare group IoU distributions with the two-sided Mann–Whitney U test. IoU distributions are non-Gaussian and heavy-tailed, so a non-parametric rank test is appropriate. Significance is declared at $p < 0.05$. We report both raw and stratified gaps.

5 Demographic-Weighted Bounding Box Regression

DWBRR applies the Group DRO reweighting principle [11] to the CIOU regression branch. Standard YOLO training minimizes $L_{\text{class}} + \lambda L_{\text{reg}}$. DWBRR replaces the regression term with a group-weighted sum:

$$L_{\text{DWBRR}} = L_{\text{class}} + \lambda \sum_i w_{g_i} L_{\text{CIOU}}(b_i, \hat{g}_i), \quad (2)$$

where $w_g \propto 1/\mathbb{E}_{\text{train}}[\text{IoU} \mid g]$ and weights are renormalized each epoch so that $\sum_g w_g = 1$. No architecture change is required. We override the Ultralytics training callback to recompute weights from per-group training CIOU and scale the regression loss before backpropagation.

Training protocol. Both DWBRR and an identically configured baseline (unweighted YOLOv8n) are trained on a 1,693-image BDD100K subset with reliable gender labels (64.8% male and 35.2% female instances), for 50 epochs, batch size 8, image size 416×416 , $\text{lr}_0 = 0.01$ with cosine decay, on a single NVIDIA RTX 3050. Validation uses a disjoint 423-image subset. We replicate with seeds $\{42, 123, 7\}$ and select the checkpoint with the best overall validation CIOU unless otherwise noted. This protocol isolates the effect of demographic regression reweighting from architecture or optimizer changes.

6 Results

6.1 Phase 1: Localization Parity Audit

Figure 1 summarizes the study design. Tables 1 and 2 and Figure 2 report the Phase 1 audit.

Gender. Female pedestrians receive lower mean IoU than male pedestrians in all 12 dataset and detector cells (mean gap -1.40 pp; 11 of 12 significant). The lone non-significant cell is CityPersons with Cascade R-CNN (-0.49 pp, $p = 0.11$). BDD100K shows the most consistent gaps (-2.16 , -2.02 , and -2.32 pp). EuroCity-Night is comparably severe.

Age. Children receive lower mean IoU than adults in all 12 cells (mean -3.27 pp; 11 of 12 significant), roughly double the gender gaps. EuroCity-Night is the harshest setting across all three detectors (-4.35 , -4.34 , and -4.44 pp).

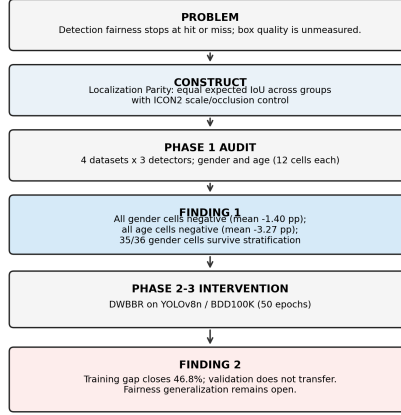


Fig. 1. Conceptual framework. Phase 1 audits Localization Parity across four datasets and three detectors under ICON² confounder control. Phase 2 and Phase 3 test whether DWBBR closes the gap and reveal a fairness generalization barrier.

Confounder control. Under ICON² scale stratification the gender gap survives in 35 of 36 cells (mean -1.43 pp, nearly identical to the unstratified -1.40 pp). The age gap survives in 24 of 30 within-scale cells. The disparities are therefore not reducible to women and children occupying smaller image regions.

6.2 Phase 2 and Phase 3: DWBBR Intervention

Training dynamics. On the training set, DWBBR behaves as intended (Figure 3): the female to male CIoU gap narrows from -2.82 pp at epoch 1 to -1.50 pp at epoch 50 (46.8% relative reduction), while the female loss weight drifts toward parity as the per-group IoU ratio improves.

Validation. On held-out data the picture changes (Table 3, Figure 4). Under the primary seed, DWBBR female CIoU is 0.7715 against baseline 0.7790; the female to male gap widens from -1.52 pp (baseline) to -2.74 pp (DWBBR). A one-sided Mann-Whitney test finds no significant female improvement ($p = 0.71$). Across three seeds the mean validation gap moves from -1.78 ± 0.48 pp (baseline) to -1.36 ± 0.63 pp (DWBBR), a 23.6% mean reduction accompanied by higher variance and unstable best-checkpoint epochs ($\{15, 17, 21\}$ for DWBBR versus $\{34, 17, 24\}$ for baseline).

Fairness generalization gap. DWBBR reliably closes the training LP gap, yet the improvement does not transfer cleanly to validation: variance rises, checkpoint selection becomes unstable, and under the primary seed the validation

Table 1. Phase 1 LP audit, gender axis. Gap = female minus male mean IoU (pp). †: $p < 0.05$, Mann–Whitney U . All 12 gaps are negative; 11 of 12 are significant.

Dataset	Detector	Gap (pp)
BDD100K	YOLOX	$-2.16^\dagger$
BDD100K	RetinaNet	$-2.02^\dagger$
BDD100K	Cascade R-CNN	$-2.32^\dagger$
CityPersons	YOLOX	$-0.98^\dagger$
CityPersons	RetinaNet	$-0.71^\dagger$
CityPersons	Cascade R-CNN	-0.49
EuroCity-Day	YOLOX	$-0.67^\dagger$
EuroCity-Day	RetinaNet	$-0.65^\dagger$
EuroCity-Day	Cascade R-CNN	$-0.98^\dagger$
EuroCity-Night	YOLOX	$-1.81^\dagger$
EuroCity-Night	RetinaNet	$-2.47^\dagger$
EuroCity-Night	Cascade R-CNN	$-1.59^\dagger$
Mean		-1.40

gap worsens. This aligns with known limits of group reweighting under train and test demographic shift [11]. We report it as a finding rather than as a claim that all mitigations must fail, and we treat fairness-aware checkpoint selection and stronger DRO-style objectives as concrete next steps. Figure 5 shows qualitative Localization Parity violations on BDD100K.

Answers to the research questions.

- **RQ1.** Yes. Female pedestrians receive lower mean IoU in all 12 cells (mean -1.40 pp; 11/12 significant), and children in all 12 cells (mean -3.27 pp; 11/12 significant). Consistency across datasets and architectures is the main signal: the disparity is systematic.
- **RQ2.** Yes. Under ICON² scale stratification the gender gap survives in 35 of 36 cells and the age gap in 24 of 30. Even among same-sized pedestrians, disadvantaged groups keep lower mean IoU.
- **RQ3.** Partially. DWBBR cuts the training gap by 46.8%, so the mechanism works where it can see the data, but the validation transfer is weak and unstable. That is the fairness generalization gap.

7 Ethical Considerations

Demographic labels in BDD100K are third-party visual inferences: binary gender, no self-identification, and undocumented annotation guidelines. These raise consent, accuracy, and inclusivity concerns [12]. LP is a proxy for box-quality fairness, not a guarantee of equal downstream risk. Nonlinear collision estimators can map equal mean IoU to unequal outcomes. Fairer localization can reduce collision risk and can also improve re-identification and surveillance. We publish

Table 2. Phase 1 LP audit, age axis. Gap = child minus adult mean IoU (pp). †: $p < 0.05$, Mann–Whitney U . All 12 gaps are negative; 11 of 12 are significant.

Dataset	Detector	Gap (pp)
BDD100K	YOLOX	$-1.92^\dagger$
BDD100K	RetinaNet	-1.64
BDD100K	Cascade R-CNN	$-2.83^\dagger$
CityPersons	YOLOX	$-2.91^\dagger$
CityPersons	RetinaNet	$-3.11^\dagger$
CityPersons	Cascade R-CNN	$-3.48^\dagger$
EuroCity-Day	YOLOX	$-4.55^\dagger$
EuroCity-Day	RetinaNet	$-2.49^\dagger$
EuroCity-Day	Cascade R-CNN	$-3.17^\dagger$
EuroCity-Night	YOLOX	$-4.35^\dagger$
EuroCity-Night	RetinaNet	$-4.34^\dagger$
EuroCity-Night	Cascade R-CNN	$-4.44^\dagger$
Mean		-3.27

Table 3. Validation CIoU by group on the BDD100K validation subset (423 images). Gap = weaker group minus reference group (pp). Multi-seed means use seeds {42, 123, 7}.

Group	Model	n	CIoU	95% CI	Gap (pp)
Male	Baseline	555	0.7942	—	—
Female	Baseline	376	0.7790	[0.768, 0.790]	-1.52
Male	DWBBR	555	0.7990	—	—
Female	DWBBR	374	0.7715	[0.760, 0.783]	-2.74
Mean female to male, baseline					-1.78 ± 0.48
Mean female to male, DWBBR					-1.36 ± 0.63

because we judge the near-term safety benefit to outweigh that dual-use risk, while acknowledging that the judgment is contestable.

The study examines only gender and age. We do not report race or skin-tone LP results here: an earlier pilot lacked sufficient labeled sample size and used categories too coarse for reliable inference. Intersectional analyses (for example, young female pedestrians) and broader attributes such as disability and body type remain open. Binary gender labels also exclude non-binary identities from the analysis. Future audits should prefer self-reported or carefully validated labels wherever ethically and legally feasible.

8 Conclusion

Localization Parity is a previously unmeasured, safety-relevant fairness criterion for pedestrian detection. Across four datasets and three detector families, fe-

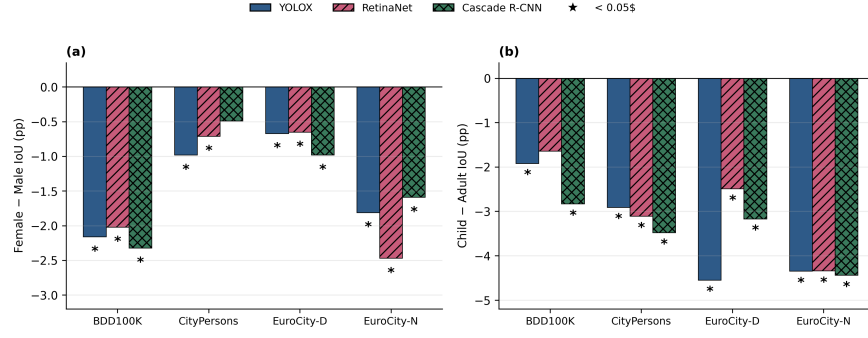


Fig. 2. Phase 1 LP audit: gender gaps (a) and age gaps (b) across four datasets and three detectors. Asterisks mark $p < 0.05$.

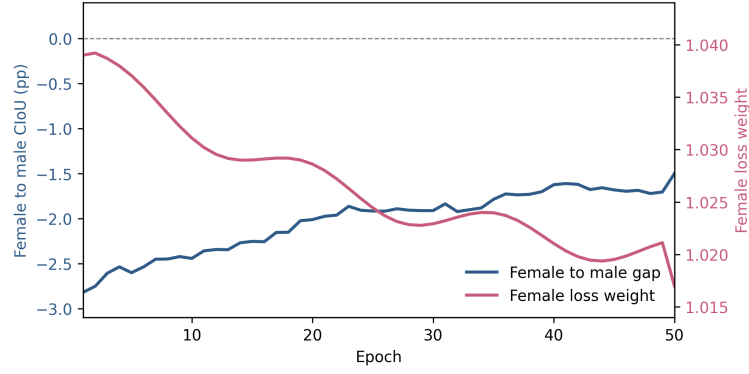


Fig. 3. DWBBR training dynamics: LP gap convergence over 50 epochs on BDD100K.

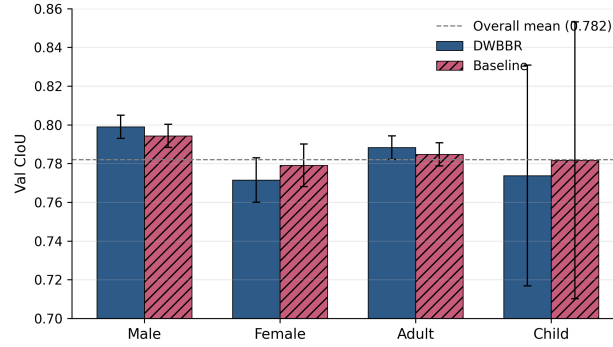


Fig. 4. Validation-set CIOU for DWBBR and the baseline, by demographic group.



Fig. 5. Qualitative Localization Parity violations on BDD100K. Female pedestrians more often receive looser predicted boxes than male pedestrians in comparable scenes.

male and child pedestrians receive systematically lower IoU boxes, and the gaps survive scale stratification. Demographic-weighted regression closes the training gap by 46.8% but does not generalize reliably. The study therefore lands two firm claims—LP is real and measurable, and it is not explained by known scale confounders—and one caution: training-time reweighting alone is not a trustworthy fix. That caution marks Localization Parity as an open research problem rather than a solved one.

8.1 Limitations

Several constraints bound the claims. First, DWBBR is trained and evaluated only on BDD100K with YOLOv8n; the fairness generalization gap might be pairing-specific or universal to reweighting—only more dataset-detector pairs would tell. Second, the validation child cohort is small ($n = 17/16$), so age-axis intervention results are statistically weak. Third, we did not log the baseline’s per-group training IoU each epoch, so the generalization finding leans on validation and multi-seed comparisons rather than an epoch-matched training curve. Fourth, inverse-proportional weighting is a heuristic borrowed from Group DRO, not proven optimal; whether the failure comes from distributional shift, overfitting, or the weighting form remains unsettled. Finally, every demographic label is a third-party visual inference, binary, and blind to non-binary or intersectional identities.

8.2 Future Work

Promising next steps include: (i) cross-dataset and cross-architecture DWBBR evaluation (train on BDD100K, test on CityPersons and EuroCity; repeat on

Cascade R-CNN); (ii) fairness-aware checkpoint selection by validation LP gap rather than mAP alone; (iii) stronger baselines such as true Group DRO on the regression loss and demographic oversampling; (iv) intersectional and skin-tone analyses once labels permit; and (v) closed-loop simulation of how 1–3 pp IoU disparities propagate into distance, time-to-collision, and braking decisions, including domains beyond autonomous driving such as surveillance and robotics.

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